

Bishoy Kamel Mamdouh AI Engineer

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Profile

- Computer Engineering student passionate about Artificial Intelligence and real-world problems. I have developed AI projects including epileptic seizure recognition using LSTM and GANs, and handwritten medical prescription data extraction using computer vision and NLP. Currently, I'm working on Retrieval-Augmented Generation (RAG) systems and AI Agents to build more reliable and context-aware AI applications.

Education

Bachelor's degree, Level 3, Computer Engineering, 08/2022 – 08/2027
Helwan University, Cairo, Egypt

Experience

Instructor, Wiro Plus 11/2024 – 09/2025

- Delivered interactive robotics sessions to primary school students, teaching basic programming, electronics, and problem-solving through hands-on projects, while fostering engagement, creativity.

Courses & Certifications

Samsung Innovation Campus Scholarship - AI Track 07/2024 – 10/2024

- Gaining practical experience in Python programming, probability, statistics, linear algebra, data analysis, web scraping, and building AI models including machine learning, deep learning, computer vision (CNNs), NLP, GANs, and model deployment.

Full Embedded Systems Diploma under supervision of Engineer Mohamed Tarek

Oral Communication B23 (CEFR), AUC - School of Continuing Education

- English Oral Communication (B2–C1 CEFR) developing strong proficiency in speaking, listening, reading, and writing for effective academic and professional communication.

Projects

RAG question answering system

- Developed a Retrieval-Augmented Generation (RAG) question answering system using FastAPI, integrating LLMs with a PostgreSQL-backed retrieval pipeline. Implemented API endpoints, managed the database with Alembic migrations, and containerized the application using Docker for production-ready deployment.

Extraction of Information from Handwritten Prescriptions

- Built using image preprocessing and OCR techniques, followed by structuring key data (drug names, dosages, instructions), and deployed the solution as an API using FastAPI.

Epileptic Seizure Recognition

- Developed using deep learning models including ANN, LSTM, and GANs, processing time-series data to detect seizure patterns.

Brain Tumor Segmentation | U-Net

- Developed using U-Net, applying Segmentation techniques to accurately segment tumor regions from medical imaging data.

Table Extraction using Yolo

- Built using YOLO to detect and localize tables in documents, followed by extracting structured data using VLM and exporting it into CSV format.

Startup Success Prediction

- Built with advanced data preprocessing, including data cleaning, feature engineering, and handling missing values, followed by training machine learning models to predict startup outcomes.

Linux-System-Programming

- Implementing multiple custom shells (Femto, Pico, Nano, and Micro Shell) and core Unix utilities, demonstrating process management, command parsing, and low-level system calls.

Tazkarti

- Developed a Bus Ticket Booking Management System as part of a Software Engineering course, applying SDLC principles, with features including authentication, booking management, seat allocation, and revenue tracking using JavaFX and MySQL.

Embedded Projects

- Dual Microcontroller-Based Door Locker Security System with Password Authentication - Car Parking Sensor - Smart Home Automation - Stopwatch System - Safe Locker Application - Voting Machine

Tools & Development Environments

Development

Visual Studio Code, Jupyter Notebook, Google Colab, Kaggle, Roboflow

Version Control

Git, GitHub

API Testing

Postman

Database

MySQL Workbench, Studio 3T

Virtualization & Environment

Docker, Docker Compose, VirtualBox, UTM

Skills

Programming Languages

Python, C++, C, Java

Core Concepts

Algorithms & Problem Solving, OOP, Debugging, Software Engineering Principles

Systems & Embedded

Linux, Shell Scripting, Microcontroller Programming, Circuit Fundamentals

AI & Machine Learning:

PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, CNNs, LSTM, GANs, NLP, RAG, AI Agents, Data Preprocessing, Feature Engineering

Databases

PostgreSQL, MySQL

MLOps & Deployment

FastAPI

Extracurricular Activities

HCC HU , AI Mentor

07/2025 – 12/2025

FEHU- CPC , Head of HR

02/2025 – 08/2025

IEEE Helwan SB , ComSoc Technical Support

09/2023 – 07/2024

Pixels Egypt Org. , Academic Projects Team

11/2023 – 07/2024

Languages

English

Advanced

Arabic

Native

French

Beginner